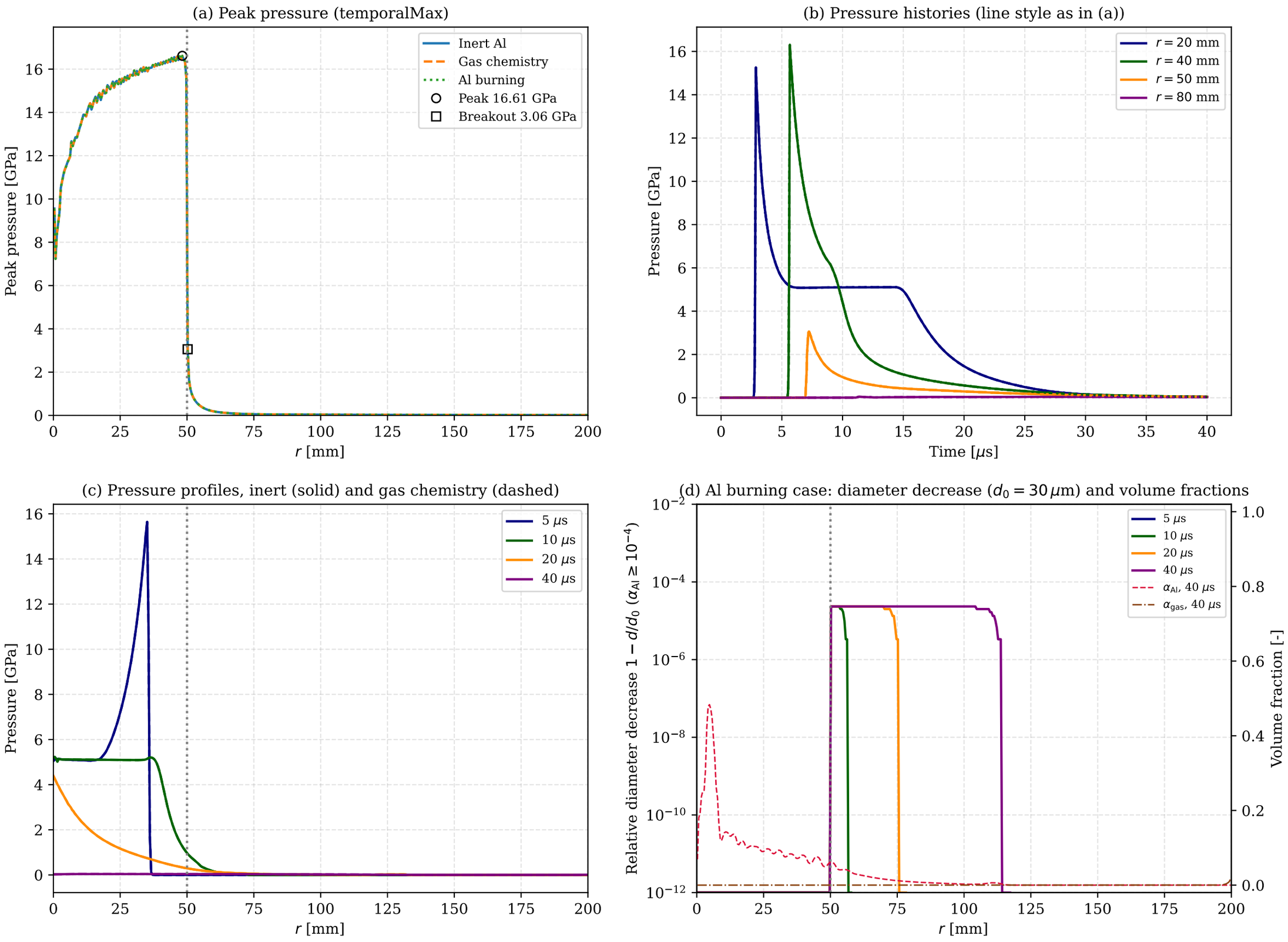
(a) Peak pressure (temporalMax)
Inert Al
Gas chemistry
Al burning
Peak 16.61 GPa
Breakout 3.06 GPa
Peak pressure [GPa]
r [mm]
(b) Pressure histories (line style as in (a))
r = 20 mm
r = 40 mm
r = 50 mm
r = 80 mm
Pressure [GPa]
Time [μs]
(c) Pressure profiles, inert (solid) and gas chemistry (dashed)
5 μs
10 μs
20 μs
40 μs
Pressure [GPa]
r [mm]
(d) Al burning case: diameter decrease ($d_0 = 30\,\mu$m) and volume fractions
5 μs
10 μs
20 μs
40 μs
α_{Al}, 40 μs
α_{gas}, 40 μs
Relative diameter decrease $1 - d/d_0$ ($\alpha_{Al} \geq 10^{-4}$)
Volume fraction [-]
r [mm]